Caroline Mazini Rodrigues, PhD

PhD in Signal, Image, Automatique

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Skills

Machine learning: Deep learning, explainable artificial intelligence (xAI), interpretability of neural networks, supervised learning, unsupervised learning, computer vision, generative AI.
Information Theory: Image Compression Techniques and Entropy-Based Models
Computational Theory and Mathematics: Signal mathematics, algorithms and complexity, rational languages' theory.
Information Retrieval: Content-based image retrieval, text and image representation.
Programming languages: Python, C / C++, Java.
Machine learning tools: Pytorch, tensorflow, hugging face, keras, captum, scikit-learn, pytorch-lightening, wandB.
Image processing tools: OpenCV, scikit-image, pillow.

Experience

Postdoctoral Researcher *IRISA – CNRS* — **France** *02/2025 –*
Conducted research on the interpretability of deep image compression networks, analyzing latent representations and entropic learning mechanisms, and developing frugal AI methods to reduce model complexity while maintaining performance.

Research and teaching associate (ATER), *Université Gustave Eiffel* — **France** *09/2023 - 09/2024*
Enhanced CNN interpretability by developing a hierarchical segmentation method to identify key image components

Research in xAI, PhD fellowship at LRE, *EPITA* — **France** *09/2020 - 08/2023*
Proposed methods in XAI for enhancing gradient-based visualization via artificial class distancing; and introducing concept decomposition for global CNN explanations, improving concept localization and enabling bias detection.

Data Scientist, *Neuralmind* — **Brazil** *02/2020 - 08/2020*
Designed deep learning applications for image and text processing, using CNNs for object detection and bounding box attribution, and Transformer models to extract contextual information from unstructured text.

Research in Complex Data Analysis, *University of Campinas (Unicamp)* — **Brazil** *02/2018 - 08/2020*
FAPESP – São Paulo Research Foundation – Master's fellowship at RECOD. Contributed to the DéjàVu big data project by developing a method to retrieve representative social media images for forensic events.

Education

PhD in Computer Science, *Université Gustave Eiffel* — **France** *09/2020 - 11/2024*
Supervisors: Professor Laurent Najman (LIGM) and Dr. Nicolas Boutry (LRE).

MSc in Computer Science, *University of Campinas (Unicamp)* — **Brazil** *02/2018 - 08/2020*
Supervisors: Professor Zanoni Dias (LOCo) and Professor Anderson Rocha (RECOD).

BSc in Computer Science, *São Paulo State University (UNESP)* — **Brazil** *02/2013 - 08/2017*
Exchange program participation: *University of Glasgow* — **Scotland** *01/2016 - 06/2016*

Publications

Complete Journal Articles

- RODRIGUES, C. M.; BOUTRY, N.; NAJMAN, L. Explaining with trees: interpreting CNNs using hierarchies. Transactions on Machine Learning Research, 2026.
- RODRIGUES, C. M.; BOUTRY, N.; NAJMAN, L. Unsupervised discovery of Interpretable Visual Concepts. Information Sciences, 2024.
- RODRIGUES, C. M.; BOUTRY, N.; NAJMAN, L. Transforming gradient-based techniques into interpretable methods. Pattern Recognition Letters, 2024.
- RODRIGUES, C. M.; SORIANO-VARGAS, A.; BAHRAM, L.; ROCHA, A.; DIAS, Z.. Manifold Learning for Real-World Event Understanding. IEEE Transactions on Information Forensics and Security. 2021.

- PADILHA, R.; RODRIGUES, C. M.; ANDALO, F. A.; BERTOCCO, G.; DIAS, Z.; ROCHA, A. . Forensic Event Analysis: From Seemingly Unrelated Data to Understanding. IEEE SECURITY & PRIVACY. 2020.

Conference Proceedings

- RODRIGUES, C. M.; KERIVEN, N.; MAUGEY, T. Compressing image encoders via latent distillation. Under review.
- DOH, M.; RODRIGUES, C. M.; BOUTRY, N.; NAJMAN, L.; MANCAS, M.; BERSINI, H. Bridging Human Concepts and Computer Vision for Explainable Face Verification. BEWARE-23 Joint Workshop @ AIxIA, 2024.
- RODRIGUES, C. M.; BOUTRY, N.; NAJMAN, L. . Gradients Intégrés Renforcés. Explain'AI Conférence Francophone sur l'extraction et la gestion des connaissances (EGC). 2023.
- RODRIGUES, C. M.; PEREIRA, L. ; ROCHA, A. R. ; DIAS, Z. . Image Semantic Representation for Event Understanding. 2019 IEEE International Workshop on Information Forensics and Security (WIFS). 2019.
- RODRIGUES, C. M.; PITERI, M. A.; ARTERO, A. O.; ELER, D. M.; SILVA, F. A.; PEREIRA, D. R. . Facial Recognition in Digital Images using Local Binary Pattern Methods. XIII Workshop de Visão Computacional (WVC). 2017. v. 1.

Awards

Best presentation – PhD day MSTIC (2021).
Academic Merit, São Paulo State University – UNESP (2017).

Events participation

Journées Filles Maths et Informatique (JFMI). 2025. Science communication for high-school girls (Speed-meeting).
Conférence IA – ISTIC. 2025. Presentation about IA for high-school girls (Presentation).
Safari des métiers – Rennes. 2025. Importance of more women in computer science (Panel).
Hi! PARIS Summer School 2025 – Institut Polytechnique de Paris. 2025. (Summer school).
Paris Open Source AI Summit. 2025. (Meeting)
MLx Generative AI – Oxford Machine Learning Summer School (OxML). 2024. (Summer school).
EuADS Data Science for Explainable and Trustworthy AI. 2023. (Summer school).
Explain'AI (EGC). Presentation "Gradients Intégrés Renforcés". 2023. (Workshop).
Oxford Machine Learning Summer School (OxML). 2022. (Summer school).
École Jeune chercheu/r/se/s en Informatique Mathématique. Presentation "Visual xAI techniques". 2022. (Summer school).
Latin American Meeting In Artificial Intelligence (KHIPU) – Presentation "Complex Data Relevance Analysis for Event Detection". 2019. (Meeting).

Teaching

Images and fundamentals – M1 Informatique – TP – 2023/2024 – UGE – 12h
Algorithms and programming – L1 Mathématiques et Informatique – TP/TD – 2023/2024 – UGE – 88h
Project – L1 Mathématiques et Informatique – TP – 2023/2024 – UGE – 14h
Databases – L2 Informatique – TP – 2023/2024 – UGE 24h
Introduction to C programming – L2 Informatique – TP – 2023/2024 – UGE 24h
Python Big Data – Master SCIA/Image+Santé – TP – 2021/2022 and 2022/2023 – EPITA – 48h
Signal mathematics – Master SCIA/Image+Santé – TP/TD – 2021/2022 and 2022/2023 – EPITA – 20h
Introduction to Neural Networks – Master SCIA/Image+Santé – TD – 2021/2022 – EPITA – 12h
Rational Languages' theory – L1 – TP/TD – 2021/2022 – EPITA – 18h
Algorithms complexity – L1 – TD – 2021/2022 – EPITA – 14h

Languages

Portuguese [Native] – **English** [Advanced] – **French** [Advanced] – **Spanish** [Basic] – **German** [Basic – Learning]